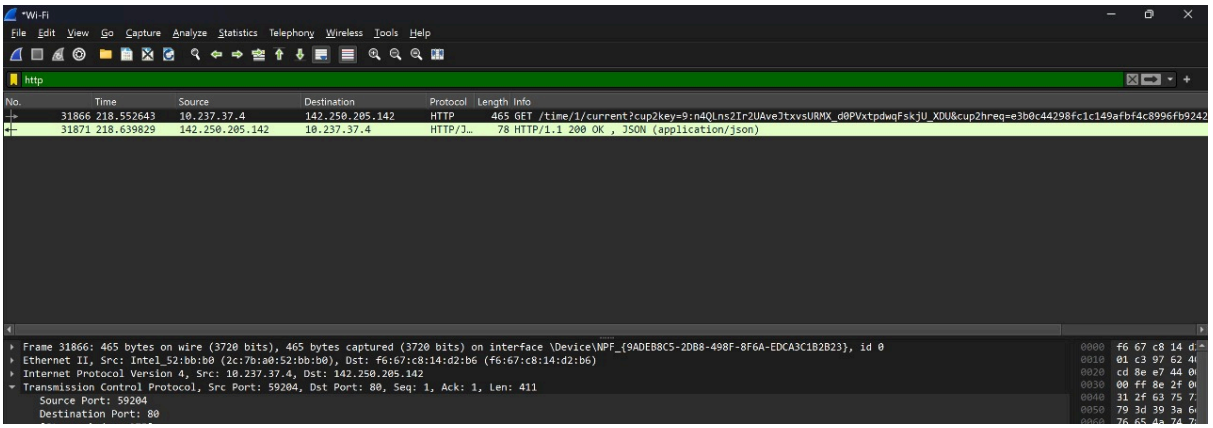
# Task 5:Capture and Analyze Network Traffic Using Wireshark:

## Monitoring Network Traffic:

## Analysis of HTTP, DNS, and TCP Protocols Using Wireshark:

### 1. HTTP (HyperText Transfer Protocol):

- **Purpose:** Used for transferring web pages and data between a web browser (client) and a web server.
- **Observation:**
  When accessing websites through Google Chrome, multiple HTTP request and response packets were captured.
  These packets showed details like the request method (GET/POST), source and destination IP addresses, and response status codes.
- **Learning Outcome:**
  Understood how web browsers communicate with servers using HTTP and how web content is transmitted across the network.

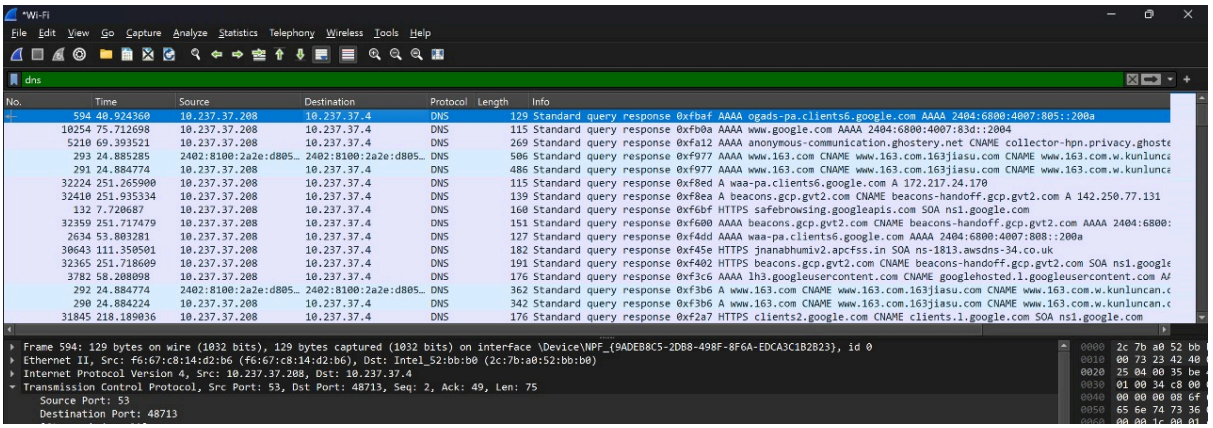



### 2. DNS (Domain Name System):

- **Purpose:** Translates human-readable domain names (like *www.google.com*) into IP addresses required for routing traffic.
- **Observation:**
  DNS query and response packets were captured each time a new website was accessed. The packets contained information such as the

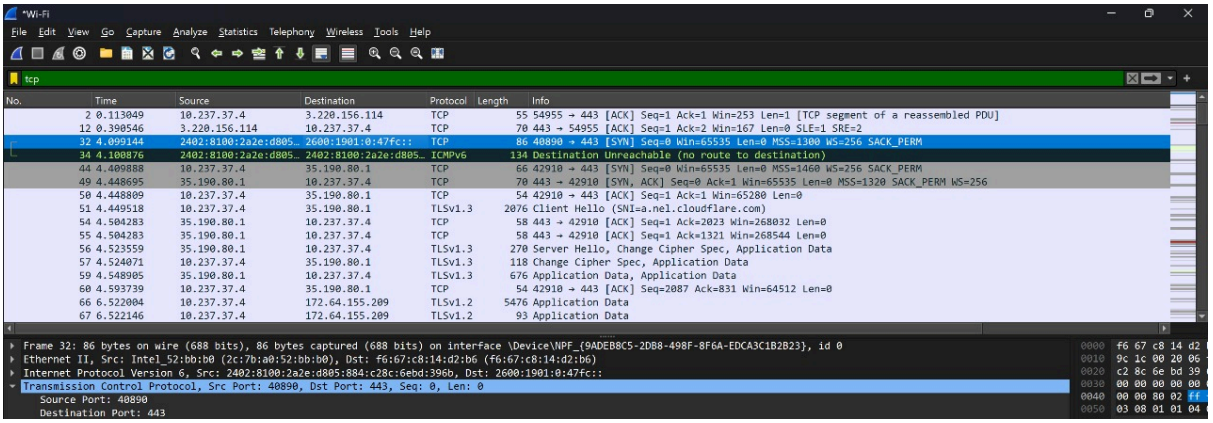
queried domain name, DNS server IP, and the resolved IP address.

- **Learning Outcome:**
  Learned that DNS is the first step in establishing most network connections, enabling devices to find the correct destination servers.




## 3. TCP (Transmission Control Protocol):

- **Purpose:** Provides reliable, ordered, and error-checked delivery of data between devices.
- **Observation:**
  Captured several TCP packets showing the three-way handshake process (SYN, SYN-ACK, ACK) during connection setup.
  Also observed data transfer and connection termination (FIN packets), confirming TCP's role in ensuring reliable communication.
- **Learning Outcome:**
  Understood how TCP ensures data integrity and connection reliability for protocols like HTTP and HTTPS.




## Wireshark Analysis Summary:

Through Wireshark analysis, I learned how different protocols interact:

- **DNS** initiates communication by resolving domain names.
- **TCP** establishes a reliable connection between devices.
- **HTTP** transmits actual web data over the established TCP connection.

This exercise enhanced my practical understanding of **network packet structure, protocol flow, and the importance of monitoring network traffic** to identify normal and abnormal behavior.